

DEBATE

FAMILY RELATIONS

Molecular analyses of the Eulipotyphla families suggest that they share a common ancestor and so form a natural group. The results also indicate that the shrews are more closely related to the hedgehogs and gymnures than to the talpids and solenodons as was previously thought. However, the relationship of the talpids to the shrews and hedgehogs and gymnures has yet to be resolved.



16–28 cm
6½–11 in

COMMON LONG-EARED HEDGEHOG *Hemiechinus auritus*

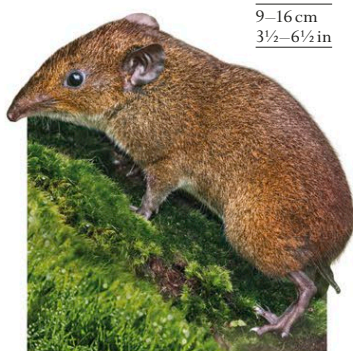
Long ears help radiate heat and keep this nocturnal hedgehog cool in the deserts of N. Africa and C. Asia.



25–46 cm
10–18 in

MOONRAT *Echinosorex gymnurus*

Resembling a large rat, the nocturnal, white-coated moonrat lives in swamps and other wet habitats in Malaysia.



9–16 cm
3½–6½ in

SHORT-TAILED GYMNURE *Hylomys suillus*

Usually solitary, this S.E. Asian gymnure lives in forested areas with dense undergrowth. It uses its long, mobile snout to find invertebrate prey but will also eat fruits.

SOLENOTODONS

The Solenodontidae are the oldest extant lineage of the group and are thought to have separated from other Eulipotyphla about 75 million years ago. They have an elongated, flexible, cartilaginous snout; a long, naked, scaly tail; small eyes; and coarse, dark fur. Unusually among mammals, they also have venomous saliva, which is used to subdue prey ranging from invertebrates to small reptiles.

27–49 cm
10½–19½ in



HISPANIOLAN SOLENOTODON *Solenodon paradoxus*

One of two extant species in its family, this solenodon is known only on Hispaniola, an island to the east of Cuba in the Caribbean.

CUBAN SOLENOTODON *Solenodon cubanus*

With longer, finer fur than its Hispaniolan relative, this secretive, nocturnal burrower was erroneously believed to be extinct during the 20th century.

20–36 cm
8–14 in



shaggy, brown coat

MOLES AND DESMANS

Small, dark insectivores with a cylindrical body, short, dense fur, and a very sensitive, hairless, tubular snout, moles of the family Talpidae are well adapted for a life of burrowing – the forelegs have powerful claws, and the hands are turned permanently outwards in the form of a shovel. In contrast, the aquatic desmans have webbed paws fringed with stiff hairs and a long, flattened tail to aid in swimming.

STAR-NOSED MOLE *Condylura cristata*

A semi-aquatic N. American mole, this species has 11 pairs of pink fleshy appendages around its snout, which detect prey by touch.



9.5–13 cm
3¾–5 in

11–16 cm
4¼–6½ in

SMALL JAPANESE MOLE *Mogera imaizumii*

This small mole from Japan is found in soft, deep soils, and is distinguished from close relatives by dental features.



10–15.5 cm
4–6¼ in

dense, waterproof coat

RUSSIAN DESMAN *Desmana moschata*

The largest member of its family, this desman's webbed hind feet and long, flattened tail are adaptations for swimming in search of food.



19–24 cm
7½–9½ in

EUROPEAN MOLE *Talpa europaea*

Secretive due to its burrowing lifestyle, this mole creates extensive networks of permanent tunnels, often marked by distinctive surface mole-hills.



11–16 cm
4¼–6½ in

EASTERN MOLE *Scalopus aquaticus*

A burrowing mole from N. America, typically in moist sandy soils, this species has ears and eyes covered by skin and fur respectively.



13–15.5 cm
5–6¼ in

PYRENEAN DESMAN *Galemys pyrenaicus*

Feeding in Pyrenean mountain streams, this desman rarely digs burrows; it shelters in rock crevices or the holes of water voles.

